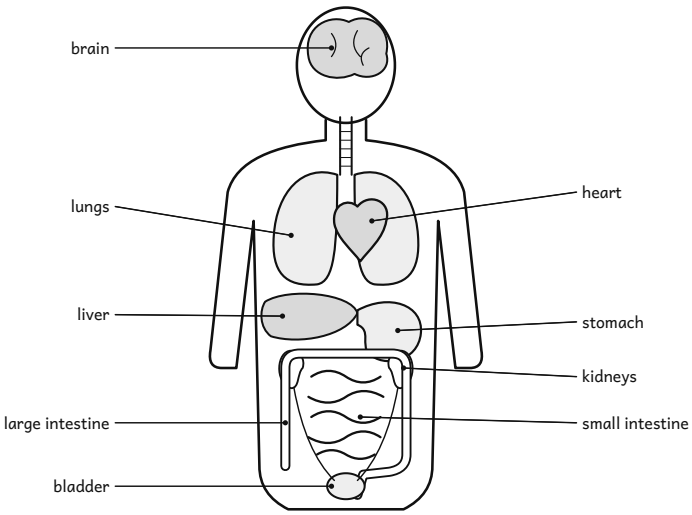
LIVING THINGS

Human Body Systems

The major organs and the digestive, respiratory, circulatory, excretory, skeletal and nervous systems



In this pack

- **Learn it:** Inside your body: organs and organ systems ★☆☆
- **Fact check:** Organs and their jobs ★☆☆
- **Think it through:** How the systems work together ★★☆
- **Investigate:** How does exercise change my pulse? ★★☆
- **Real-world case study:** Dhaka's air and our lungs ★★★
- **Review and challenge:** Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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Inside your body: organs and organ systems

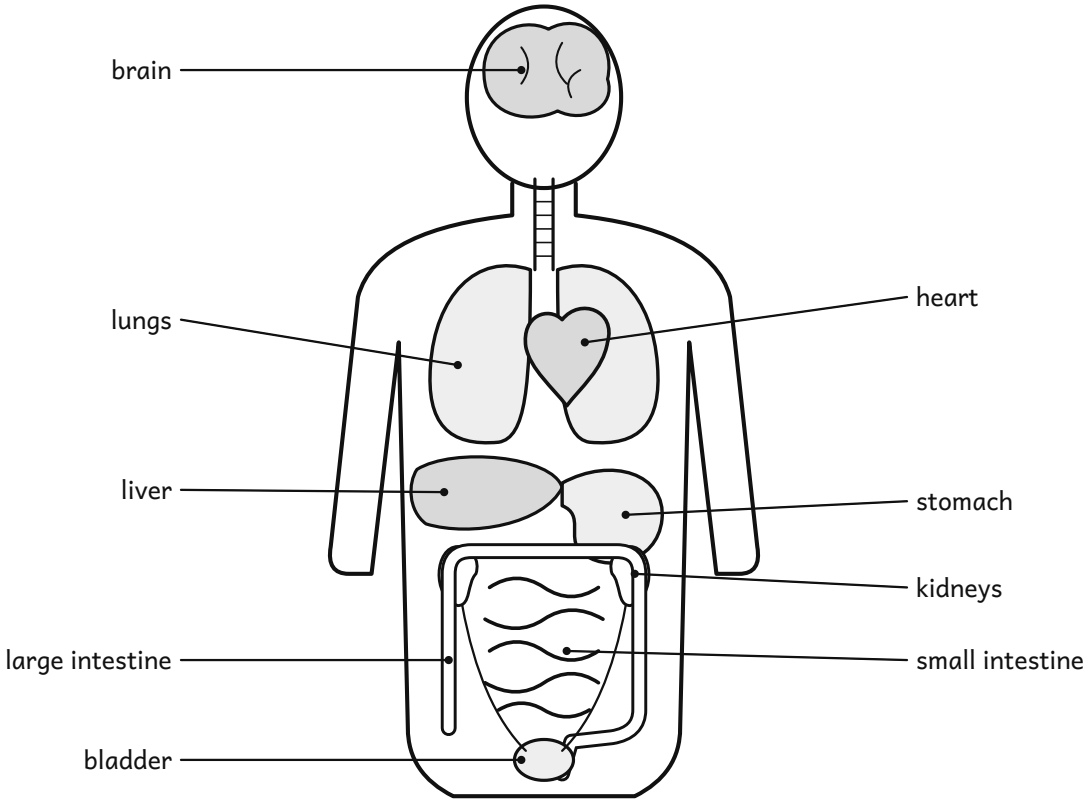
★☆☆
Grade 5

Name: Class: Date:

Learning goals

- Name the major organs, find them in the body and say what they do.
- Describe the six main organ systems.
- Explain how organ systems work together to keep us alive.

Your body is busy every second, even while you sleep. Your heart beats about 100 000 times a day, your lungs breathe in and out about 20 000 times, and your brain controls it all. These jobs are done by **organs**. An organ is a body part with a special job, such as the heart or the stomach. Organs that work together form an **organ system**.



Major organs of the human body (the kidneys are at the back).

1. Digestive system: food into fuel

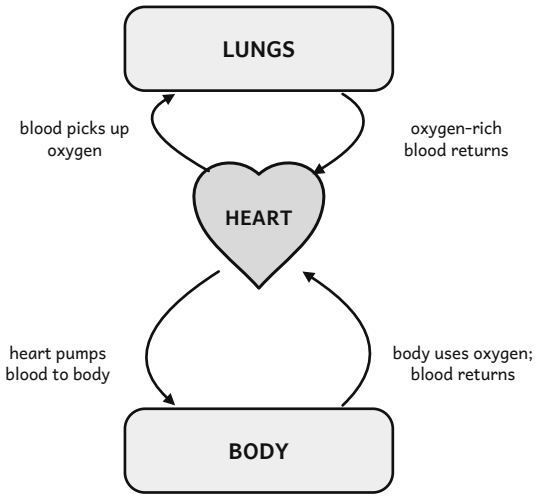
Food is broken down into tiny pieces (**nutrients**) that the body can use. Teeth chew food in the **mouth**. The food pipe (**oesophagus**) carries it to the **stomach**, where acids churn it into a soup. In the **small intestine** the nutrients pass into the blood. The **large intestine** takes back water, and waste leaves the body. The **liver** helps digestion and cleans the blood.

2. Respiratory system: breathing

Air enters through the nose and mouth and travels down the **windpipe** (trachea) into the two **lungs**. In the lungs, **oxygen** passes into the blood and **carbon dioxide** (a waste gas) passes out to be breathed out. A sheet of muscle under the lungs, the **diaphragm**, helps you breathe.

3. Circulatory system: transport

The **heart** is a muscle that pumps **blood** through tubes called **blood vessels**: arteries, veins and tiny capillaries. Blood carries oxygen and nutrients to every cell and takes waste away. Your **pulse** is the beat of blood being pushed through an artery.



The heart pumps blood to the lungs and then around the body.

4. Excretory system: removing waste

Cells make waste. The two **kidneys** filter the blood and make **urine**, which is stored in the **bladder**. The lungs remove carbon dioxide and the skin removes some waste in sweat.

5. Skeletal system: support and protection

An adult skeleton has 206 **bones**. Bones hold the body up, **protect** soft organs (the skull protects the brain; the ribs protect the heart and lungs) and work with **muscles** so we can move. Bones meet at **joints**.

6. Nervous system: control

The **brain**, the **spinal cord** and the **nerves** carry messages around the body at high speed. Sense organs (eyes, ears, nose, tongue, skin) send information to the brain; the brain decides what to do and sends messages to the muscles.

Think like a scientist

Systems work together. When you run, your brain sends messages to your muscles, your lungs breathe faster to take in more oxygen, and your heart beats faster to carry it to your muscles.

Did you know?

Laid out flat, the inside of your small intestine would cover about 30 square metres, roughly half a badminton court, because it is lined with millions of tiny finger-like folds that soak up food.

Key words

organ	a body part with a special job
organ system	organs working together
nutrients	useful substances from food
oxygen	gas from the air that cells need
carbon dioxide	waste gas we breathe out
blood vessel	tube that carries blood
pulse	the beat of blood in an artery
excretion	getting rid of waste made by the body
joint	where two bones meet

Organs and their jobs

★☆☆
Grade 5

Name: Class: Date:

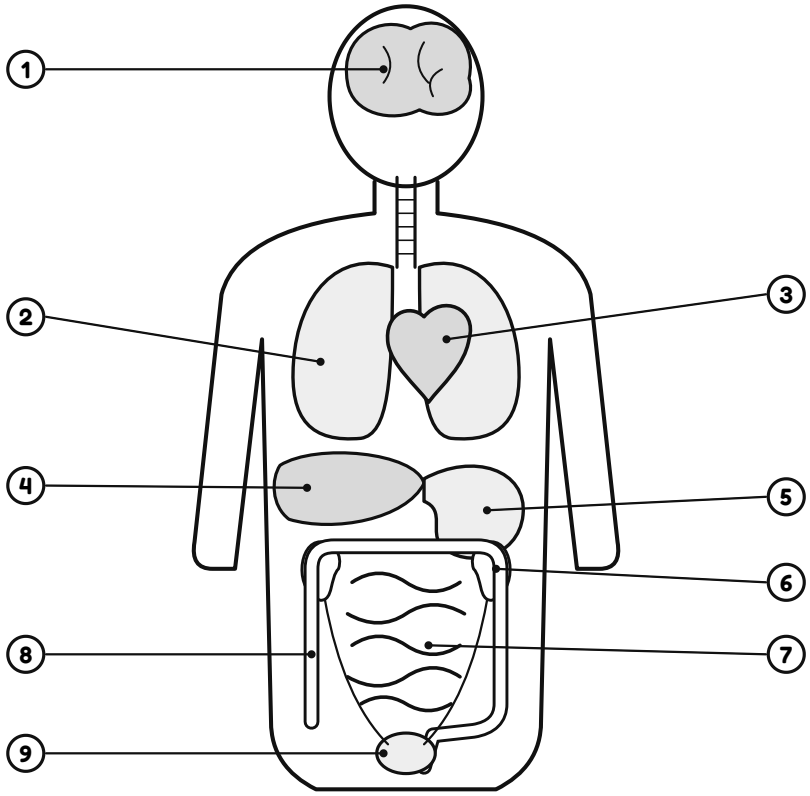
Learning goals

- Label the major organs.
- Match organs to systems and jobs.

1 Label the organs. [9]

LABELS

liver **small intestine** **stomach** **brain** **bladder** **lungs** **heart** **kidneys** **large intestine**



1 _____ 2 _____ 3 _____

4 _____ 5 _____ 6 _____

7 _____ 8 _____ 9 _____

2 **Write the organ system each organ belongs to.** [6]

Organ	Organ system
heart	
lungs	
stomach	
kidneys	
brain	
skull	

3 **Match each organ to its job.** [6]

Draw a line to match each pair, or write the letter.

heart ●	● **A** nutrients pass into the blood
lungs ●	● **B** filter blood and make urine
small intestine ●	● **C** protect the heart and lungs
kidneys ●	● **D** controls the body and processes messages
brain ●	● **E** take in oxygen and remove carbon dioxide
ribs ●	● **F** pumps blood around the body

4 **Fill in the gaps.** [4]

WORD BANK

blood vessels **bladder** **stomach** **diaphragm** **pulse** **nerve**

Blood is carried in tubes called _____. The beat you can feel in your wrist is your _____. Urine is stored in the _____. The _____ is a muscle that helps us breathe.

THINK IT THROUGH · A1 · BODY SYSTEMS

How the systems work together

★★☆
Grade 5

Name: **Class:** **Date:**

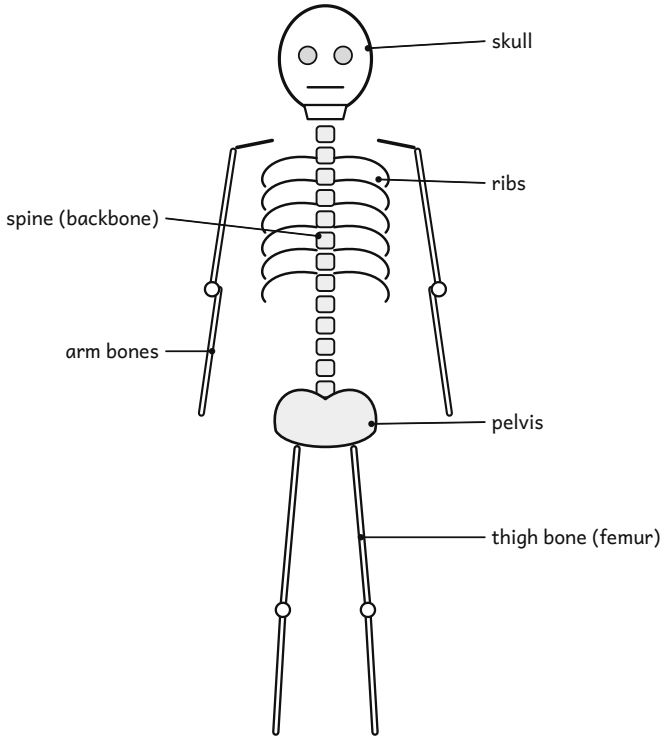
Learning goals

- Order the journey of food through the body.
- Explain links between organ systems.
- Interpret pulse data.

1 **Number the stages of digestion from 1 to 6.** [3]

- ☐ Water is taken back in the large intestine.
- ☐ Teeth chew food in the mouth.
- ☐ Waste leaves the body.
- ☐ Nutrients pass into the blood in the small intestine.
- ☐ Food travels down the oesophagus.
- ☐ The stomach churns food with acid.

2 **Which bones protect which organs? Complete the table.** [3]



Bone	Organ it protects
skull	
ribs	
spine (backbone)	

3 **Explain why the circulatory system and the respiratory system must work together.** [2]

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4 **A boy touches a hot pan and quickly pulls his hand away. Describe the path of the message.** [2]

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5 **Why do you breathe faster and your heart beat faster when you exercise?** [2]

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INVESTIGATE · A1 · BODY SYSTEMS

How does exercise change my pulse?

★★☆
Grade 5

Name: Class: Date:

Learning goals

- Measure pulse rate accurately.
- Present results in a bar chart.
- Explain a pattern using body systems.

You will need

a stopwatch or clock with a second hand, space to exercise safely

Remember

Find your pulse on the inside of your wrist, below your thumb, using two fingers (not your thumb). Count the beats for **15 seconds**, then multiply by 4 to get **beats per minute**.

1 **Practise the maths. Work out each pulse rate in beats per minute.** [3]

Beats counted in 15 s	× 4	Beats per minute
20	× 4	
23	× 4	
31	× 4	

2 **Arif measured his pulse after different activities. He measured each one three times.**

Activity (1 minute)	Test 1	Test 2	Test 3	Average
sitting still	80	76	78	
walking	96	100	98	
jogging	120	124	122	
skipping fast	148	152	150	

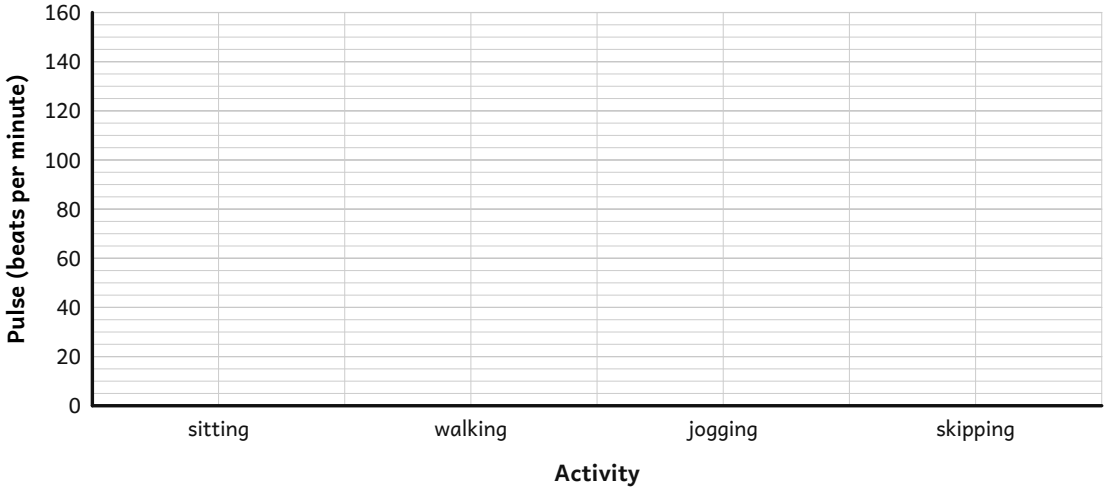
a) Calculate the averages and write them in the table.

b) Why did Arif repeat each measurement three times?

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3 **Draw a bar chart of the average pulse rates.** [4]



4 **Describe the pattern and explain it.** [2]

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5 **What could Arif investigate next? Write a question he could test.** [1]

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REAL-WORLD CASE STUDY · A1 · BODY SYSTEMS

Dhaka's air and our lungs

★★★ Grade 5

Name: **Class:** **Date:**

Learning goals

- Use data to judge air quality.
- Explain how air pollution affects body systems.
- Consider different points of view and suggest actions.

Science · Health · Maths · Geography · Citizenship

Dhaka is one of the busiest cities in the world, and in winter it is often listed among the cities with the **most polluted air**. Smoke from brick kilns, exhaust from vehicles, dust from building sites and burning rubbish add tiny particles to the air. The smallest, called **PM2.5**, are about 30 times thinner than a human hair. They travel deep into the lungs and some even pass into the blood.

Doctors say that breathing polluted air for a long time can cause coughing, asthma attacks and lung infections, and can strain the heart. Children breathe faster than adults, so they take in more polluted air for their size.

Scientists use an **Air Quality Index (AQI)** to tell people how clean the air is.

AQI categories follow the widely used US EPA scale.

AQI	0–50	51–100	101–150	151–200	201–300	301+
Air quality	Good	Moderate	Unhealthy for sensitive groups	Unhealthy	Very unhealthy	Hazardous

1 **A school in Dhaka recorded the AQI each morning for one week in January.** [4]

Day	Sun	Mon	Tue	Wed	Thu
AQI	168	212	245	186	140

a) On which day was the air the most polluted? What was its category?

b) On how many days was the air "unhealthy" or worse (151 or more)?

c) Suggest why the head teacher might cancel outdoor sports on Tuesday.

2 **Which two organ systems are harmed most by air pollution? Explain.** [2]

3 **Complete the table to show different points of view.** [3]

Person	What they might say
brick kiln owner	
a parent of a child with asthma	
a city traffic officer	

4 **Suggest three actions that could reduce air pollution or protect people's lungs.** [3]

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Show what you know

★★★
Grade 5

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the key words.**

S	T	O	M	A	C	H	H	F	S	C
O	L	U	N	G	S	N	P	B	J	B
I	N	E	R	V	E	W	J	B	N	H
F	M	K	I	D	N	E	Y	C	K	B
V	E	E	P	B	I	E	V	R	P	O
P	A	G	L	E	B	L	L	U	U	N
H	E	A	R	T	B	L	I	N	L	E
P	T	P	J	E	S	R	O	V	S	T
I	E	A	M	W	E	Y	A	O	E	A
I	T	K	V	H	T	N	I	I	D	R
W	M	E	E	E	D	M	D	F	N	A

blood bone brain heart kidney liver lungs nerve pulse stomach

2 **Which organ system includes the windpipe and the lungs?** [1]

- ◯ digestive
- ◯ respiratory
- ◯ nervous
- ◯ skeletal

3 **Which organ filters the blood to make urine?** [1]

- ◯ liver
- ◯ bladder
- ◯ kidney
- ◯ heart

4 **Sort the organs into the correct systems.** [4]

SORT THESE

spinal cord heart small intestine stomach nerves oesophagus brain blood vessels

Digestive	Circulatory	Nervous

5 **Challenge. A person can live with one kidney but not without a heart. Explain why.** [2]

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How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
name and find the major organs of the body	○	○	○
say what each major organ does	○	○	○
describe the six main organ systems	○	○	○
explain how systems work together when I exercise	○	○	○
measure my pulse and calculate beats per minute	○	○	○
explain how air pollution harms the lungs and heart	○	○	○